
Data Engineering Final Project

Proposal Due Date: 03/19/2026

Final Presentation Due Date: 03/26/2026

Table of Contents

<i>Project Objective</i>	<i>3</i>
<i>Pipeline Requirements</i>	<i>3</i>
<i>Scheduling and Monitoring</i>	<i>4</i>
<i>Choosing Your Data</i>	<i>4</i>
<i>Project Timeline</i>	<i>5</i>
<i>Final Advice</i>	<i>5</i>

PROJECT OBJECTIVE

Each individual must come up with a **business idea or use case** that involves collecting and analyzing data.

Based on that idea, you will:

- Identify and collect relevant datasets
- Build a data pipeline in Foundry
- Clean and transform the data
- Produce meaningful final outputs
- Visualize the results in an interactive dashboard

PIPELINE REQUIREMENTS

Your project pipeline should include the following components:

First, you must **ingest the datasets** you selected.

Then, you will implement **data cleaning and transformation steps** to prepare the data for analysis.

The processed datasets should then be used to generate **final outputs** that support your business idea.

These outputs must be consumed in a **dashboard** that includes:

- appropriate visualizations
- useful filters
- interactive actions for users

SCHEDULING AND MONITORING

Your pipeline should not run only once.

You must configure a **schedule** for the pipeline depending on what makes sense for your project.

For example:

- daily updates
- weekly updates
- or another schedule that fits your use case.

Additionally, you must implement **health checks** on important datasets to ensure data quality and pipeline reliability.

CHOOSING YOUR DATA

You are responsible for finding the datasets that support your project idea.

I recommend using platforms such as **Kaggle**, which contains many useful public datasets.

Before starting implementation, you must send me your proposed topic and the datasets you plan to use.

I will review them to make sure:

- the datasets are suitable
- the scope of the project is appropriate
- the topic is feasible

So please **reach out to me** with your project idea and datasets before you start building the pipeline.

PROJECT TIMELINE

Next week, each group will present a **project proposal**.

During that presentation you should explain:

- your business idea
- the datasets you selected
- the pipeline you plan to build
- the outputs and dashboard you aim to create

The following week, you will present the **final implementation** of your project.

During that presentation you should demonstrate:

- the working pipeline
- the transformations you implemented
- the scheduled pipeline
- the health checks you configured
- the final dashboard and insights

FINAL ADVICE

The goal of this project is not just to produce charts.

The goal is to demonstrate that you understand **how to design and build a complete data pipeline** that transforms raw data into useful insights.

Focus on:

- clear pipeline structure
- meaningful transformations
- useful visualizations
- good data quality practices